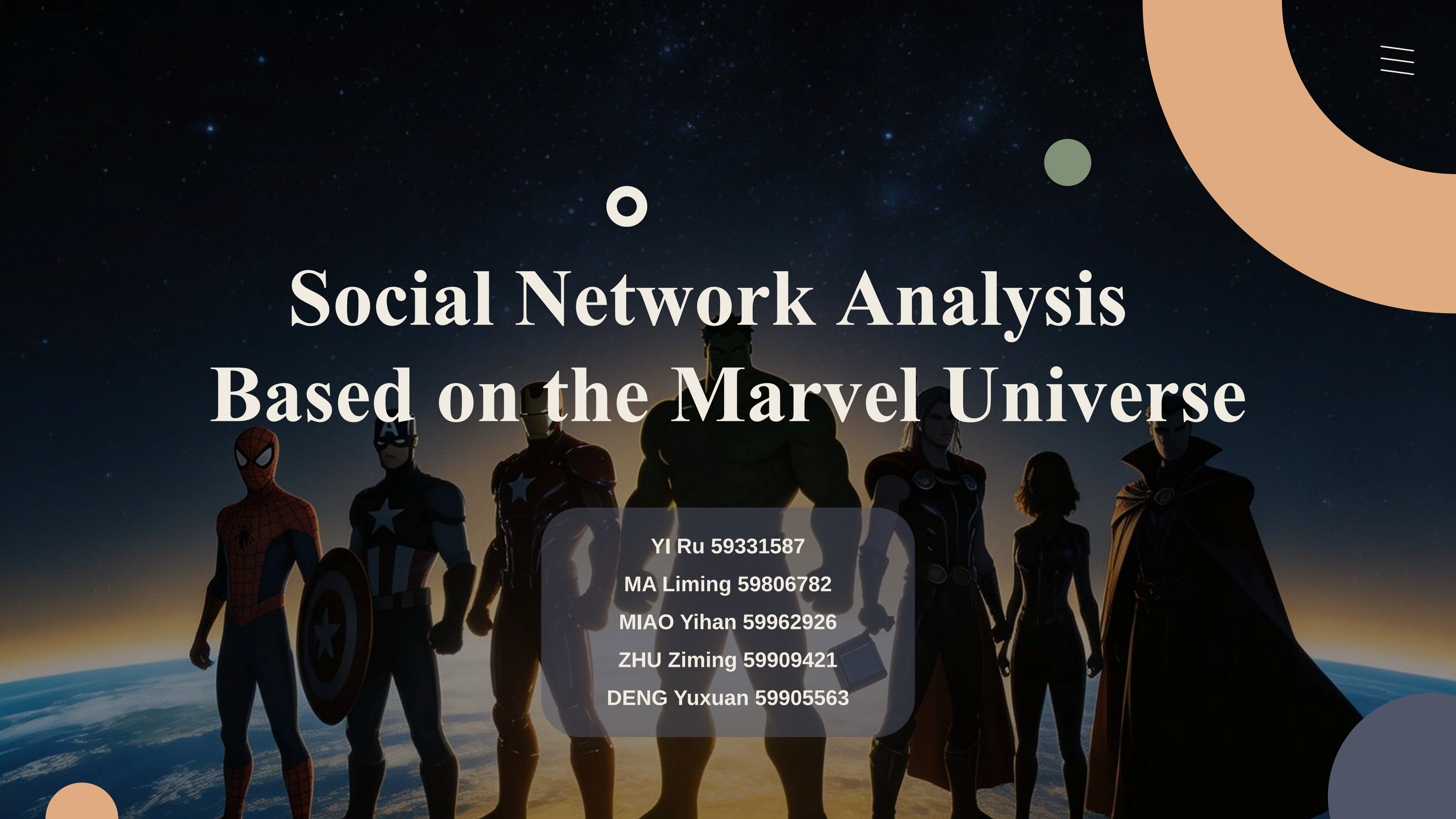




Social Network Analysis Based on the Marvel Universe

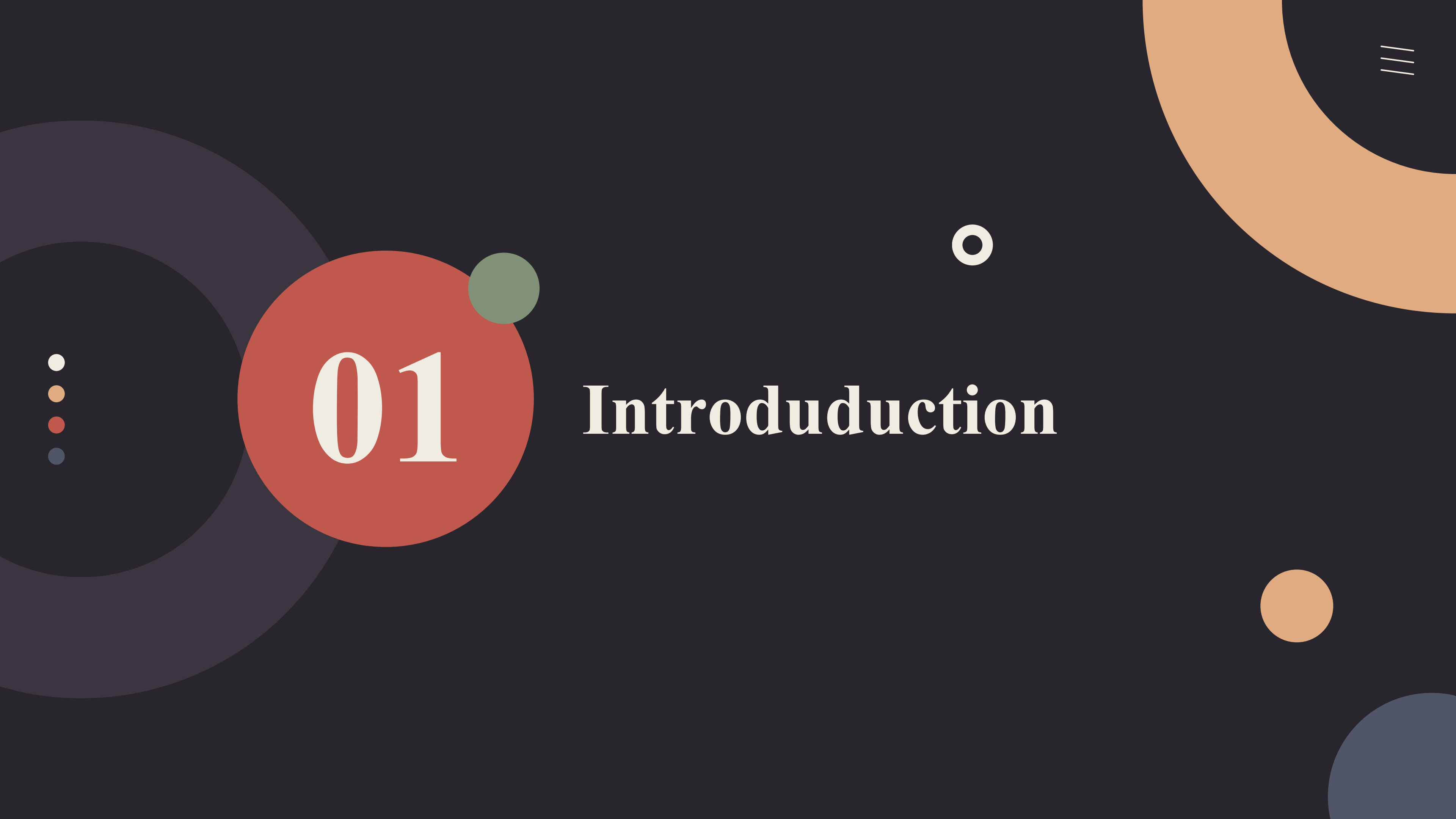
YI Ru 59331587

MA Liming 59806782

MIAO Yihan 59962926

ZHU Ziming 59909421

DENG Yuxuan 59905563



01

Introduction

Background & Significance

Why the Marvel Universe? An Ideal Research Field

- **A Vast and Complex Narrative System:** It offers a natural "laboratory" for studying large-scale, dynamically evolving social networks.
- **Dual Influence in Business and Culture:** Analyzing its network structure reveals the narrative logic behind its success, providing insights for content design.



Research Questions

- Which characters occupy the most influential network positions?
- Is there a link between a character's attributes (e.g., gender, faction) and their network centrality?
- How does the number of teams a character joins affect their structural role?
- How has the Marvel character network evolved across different historical eras?

Previous Studies

Previous Work:

- Alberich (2002):
Pioneering static analysis, but ignored time and attributes.
- Richter (2016):
Theoretical analysis, but lacked quantitative data.

Our Contribution:

- Dynamic Analysis: Examining network evolution over time.
- Multi-dimensional Attributes: Integrating gender, race, etc.
- Empirical Support: Combining theory with quantitative data.





02

Methodology

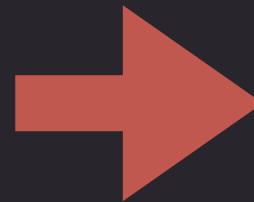
Data Source & Filtering

Data Source

- Marvel character network dataset
- Based on the Marvel Chronology Project
- Originally compiled by Rosselló, Alberich, & Miro

Original Dataset

- 19,090 nodes (characters + comic books)
- 96,662 edges (co-appearance relationships)
- Weighted & undirected network



Data Processing & Filtering

- Extracted character–character network
- Filter criteria:
 - ≥ 8 connections
 - ≥ 5 comic appearances

Final Dataset

- 327 key characters
- 9,891 edges
- Focus on most active & structurally relevant nodes

Methods & Tools

- Data processing: EXCEL, Python (pandas, networkx)

Library Name	Purpose
pandas	Data reading, cleaning, integration, and management
<u>networkx</u>	Network construction, centrality metric calculation (degree centrality, <u>betweenness centrality</u> , closeness centrality, eigenvector centrality, clustering coefficient)
<u>matplotlib</u> / <u>seaborn</u>	Data visualization (centrality distribution plots, box plots, etc.)
<u>numpy</u>	Numerical computation and array operations

- Visualization: Gephi
- Community detection: Louvain algorithm

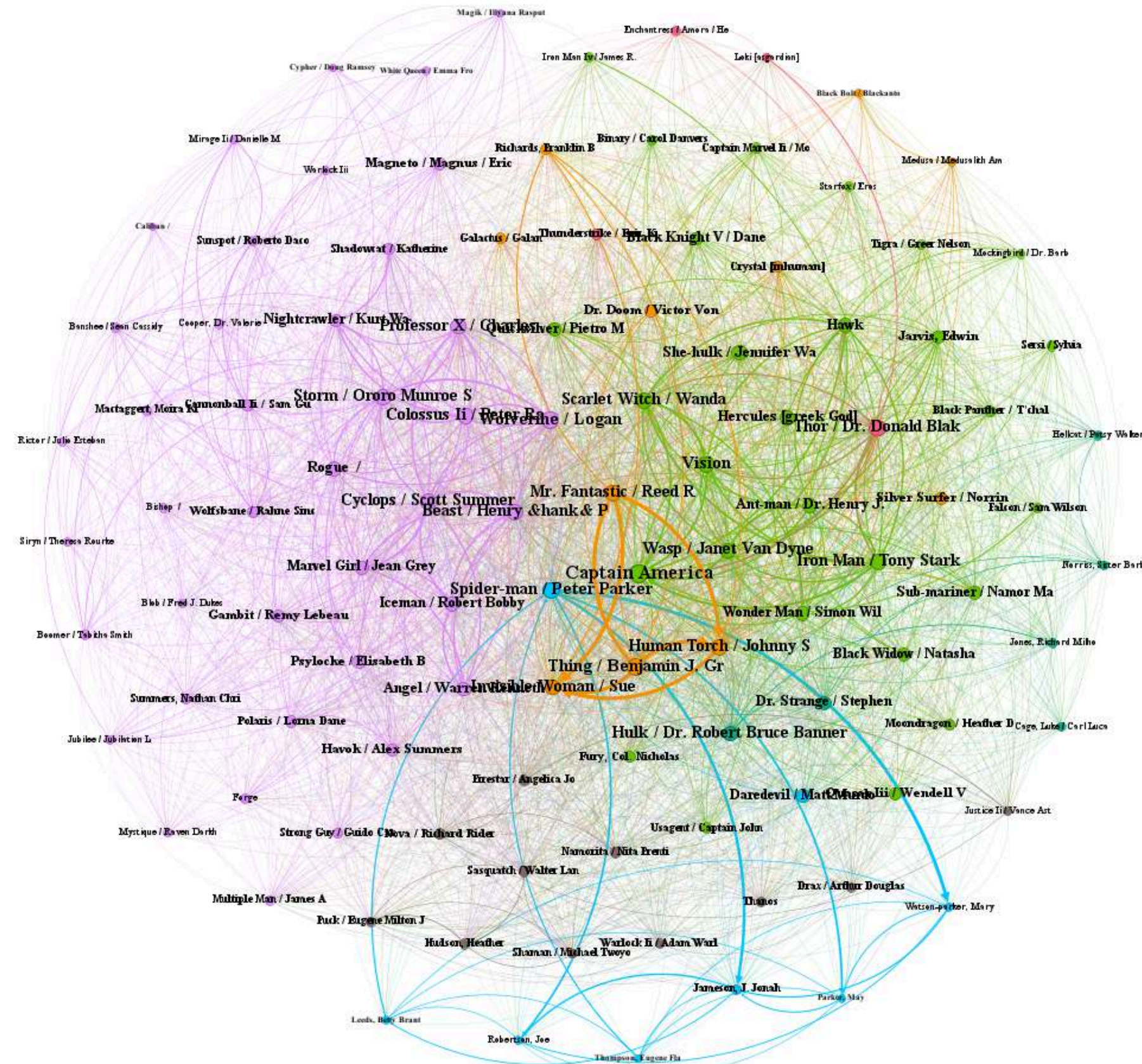




03

Results and Findings

Overall Network Structure



Fully connected network (no isolated nodes)

Small-world property:

- Average path length ≈ 1.87
- Clustering coefficient ≈ 0.476

Interpretation

Short paths $\rightarrow$ strong global connectivity

High clustering $\rightarrow$ tight local groups

Community Detection & Structure

Method

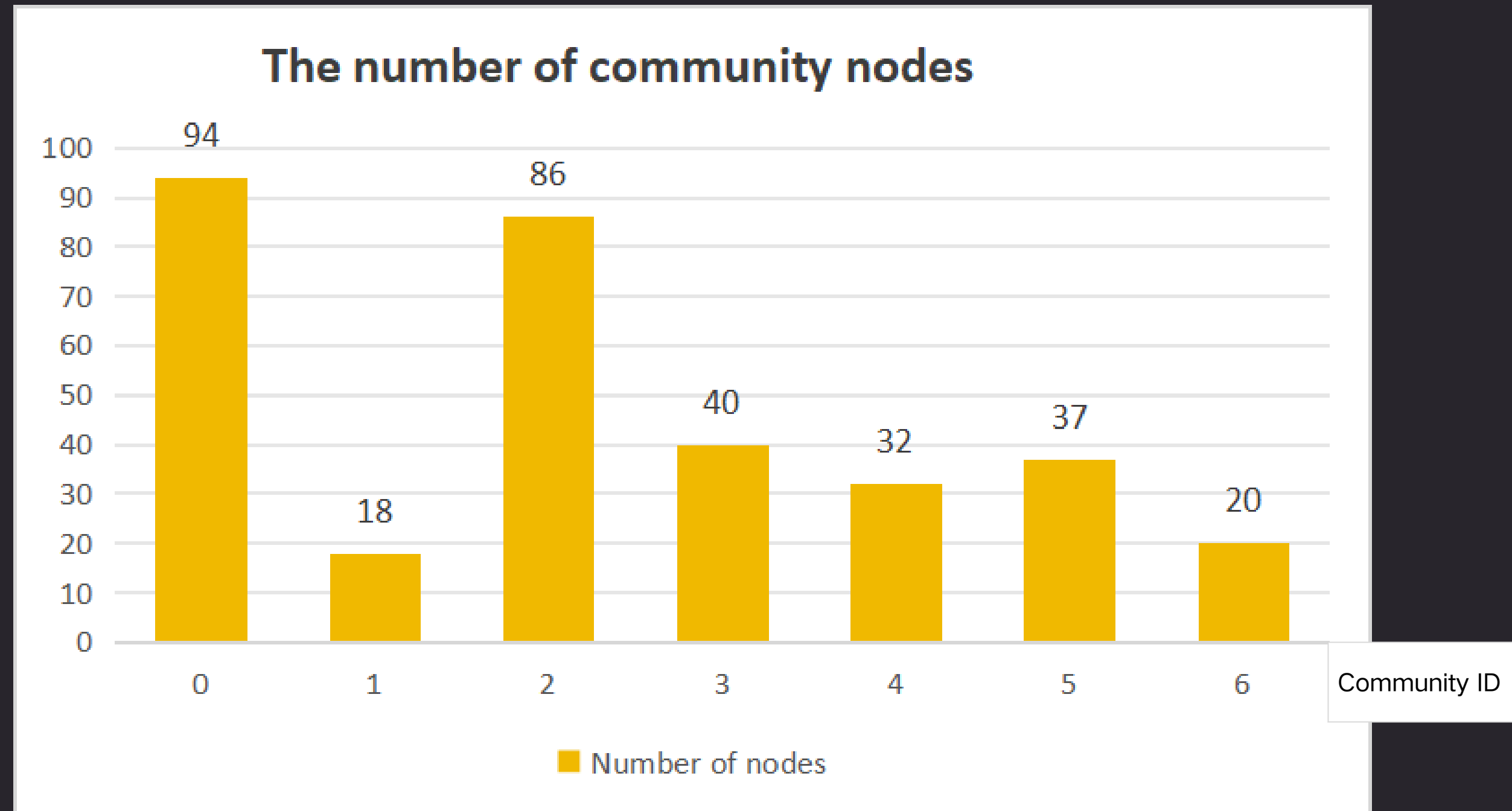
- Louvain algorithm

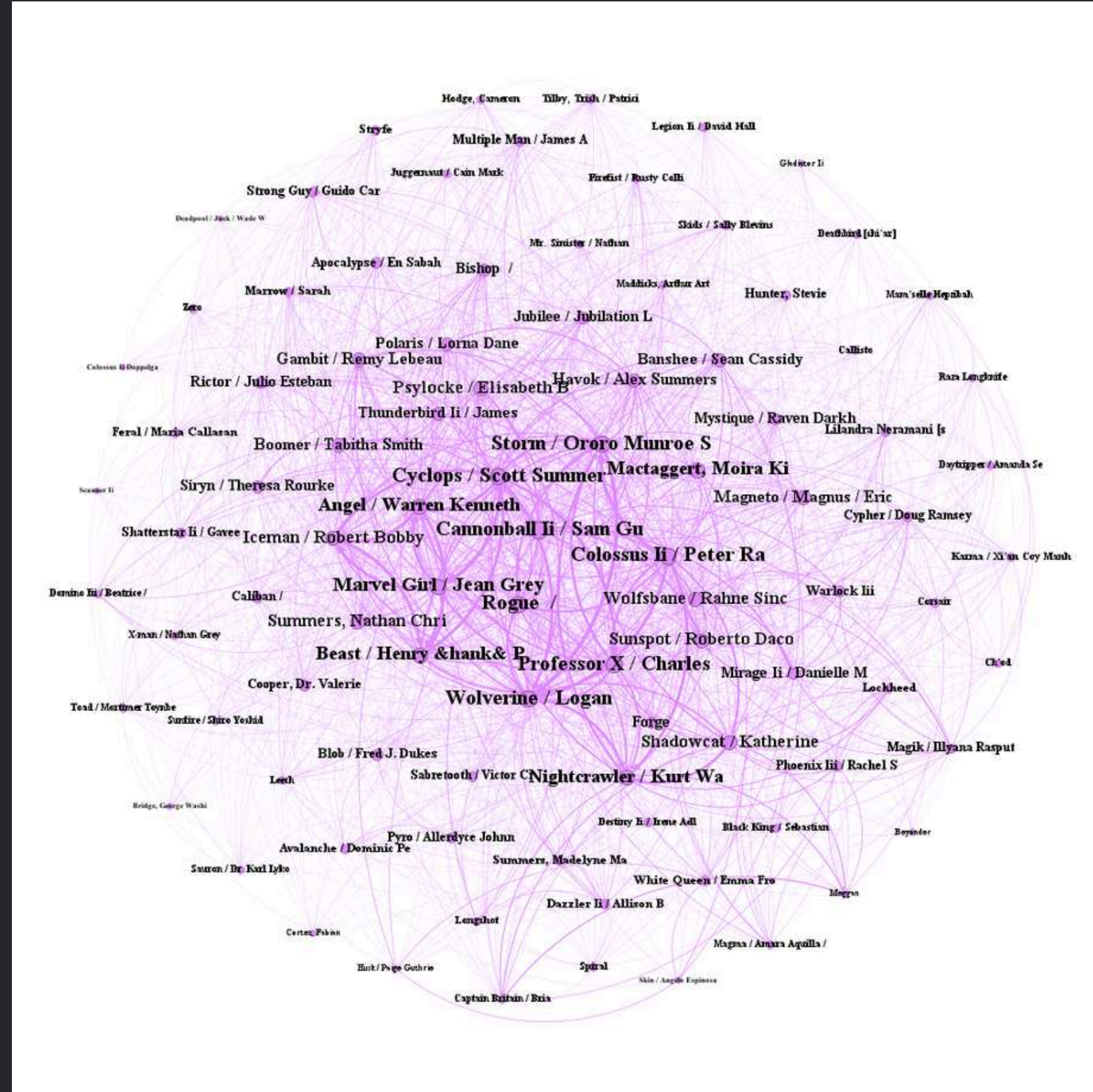
Results

- 7 communities
- Modularity: $Q = 0.435$

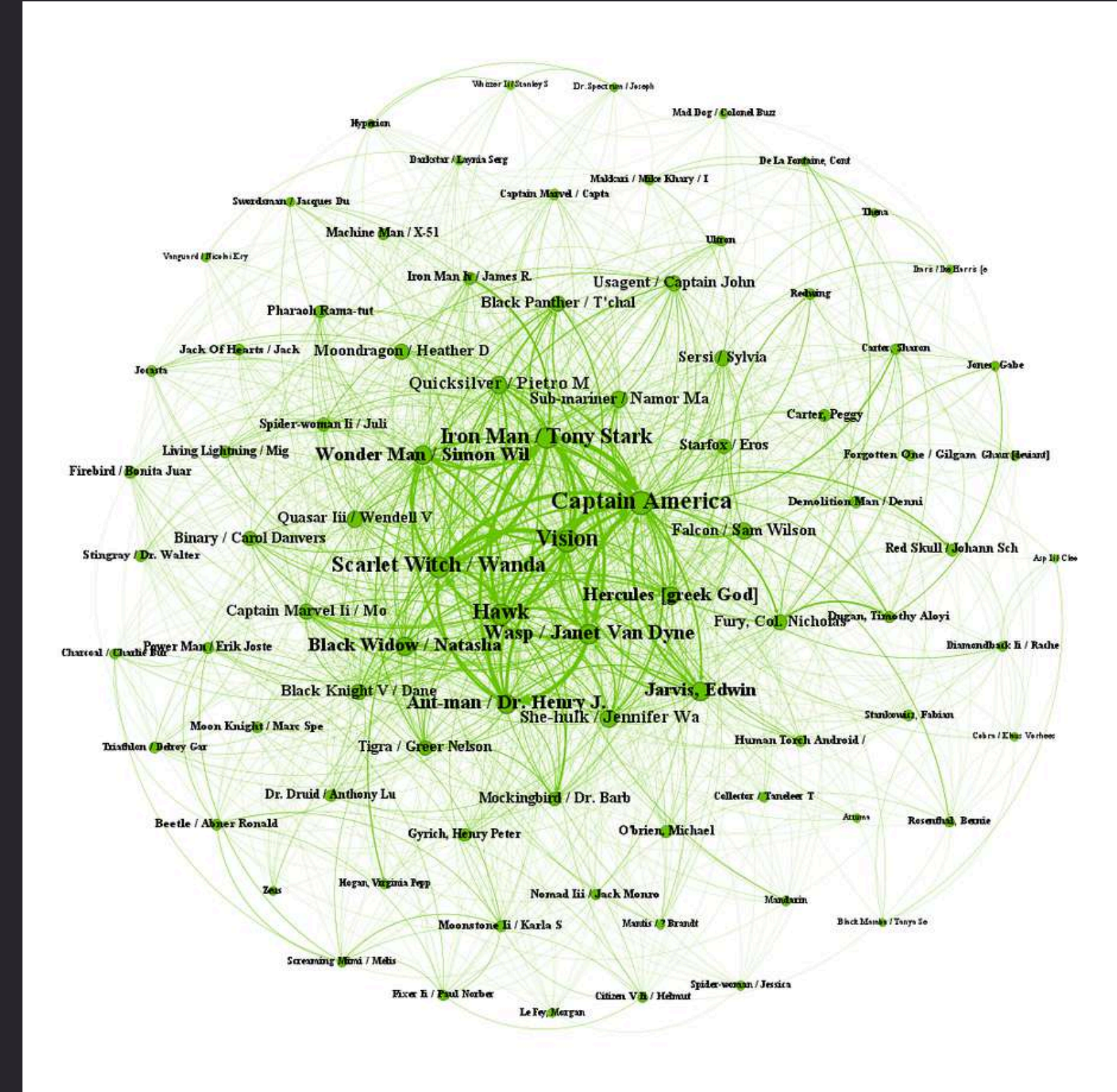
Distribution

- Largest: 94 nodes
- Second: 86 nodes
- Top 2 > 55% of network



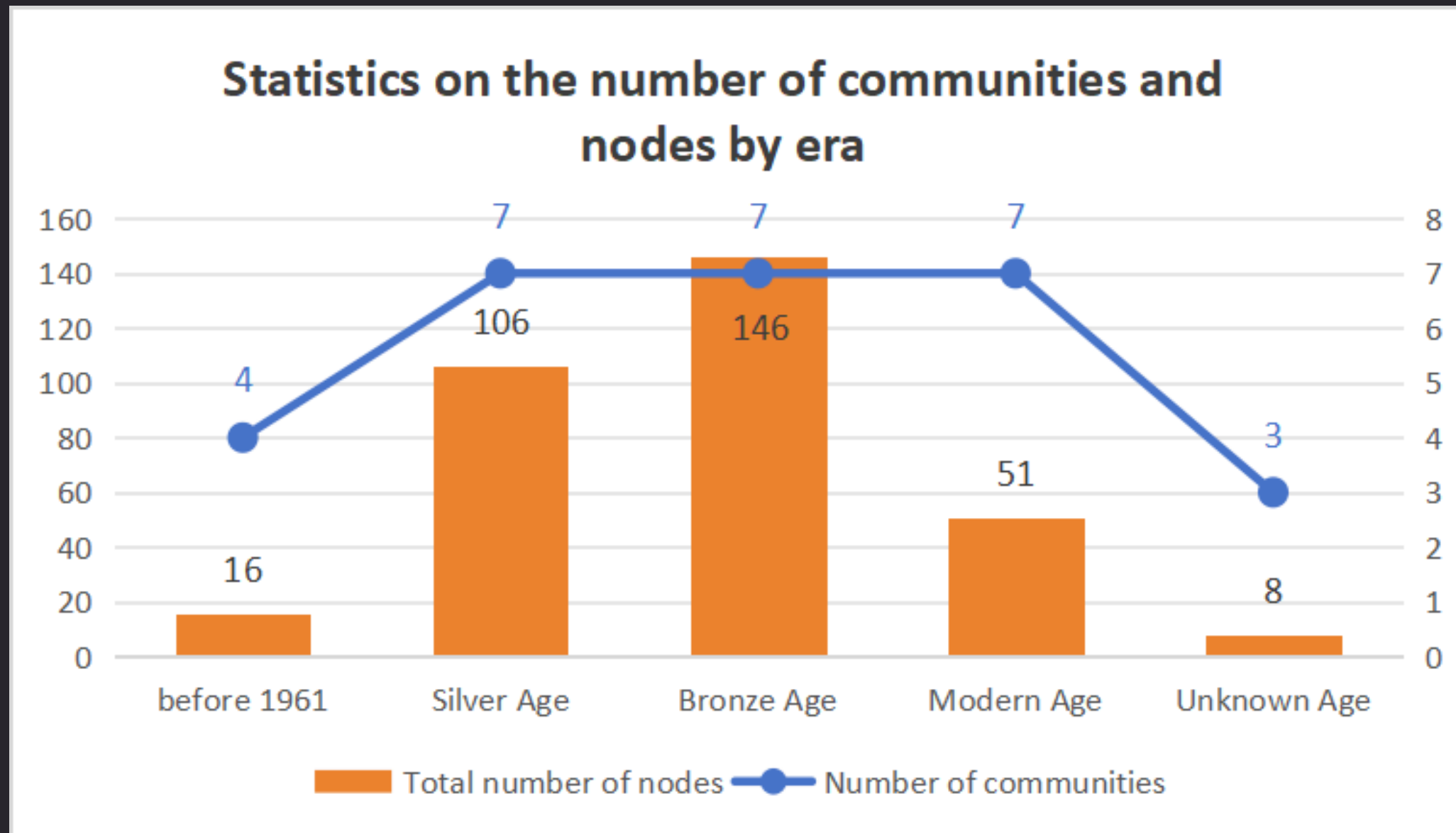


Community 0 Visualization
(Predominantly X-Men Characters)



Community 2 Visualization
(Predominantly Avengers Characters)

Temporal Evolution



- The number of characters increases significantly from the early stage to the Bronze Age, reaching a peak of 146 nodes.
- At the same time, the network quickly expands to cover all 7 communities, indicating a fully developed structure.
- In the Modern Age, the number of nodes decreases to 51, but the community structure remains stable.
- The early stage (before 1961) shows a small and incomplete network, covering only 4 communities.



Structural Change Over Time

Key Trends

- Centrality ↓ over time
- Clustering ↑ over time

Examples

- Degree: 0.30 → 0.12
- Clustering: 0.63 → 0.79

Interpretation

- Early characters dominate network
- Modern characters form smaller, tighter groups

Mean Centrality Measures by Era				
Metric	Before 1961	Silver Age	Bronze Age	Modern Age
Degree centrality	0.30	0.23	0.16	0.12
<u>Betweenness centrality</u>	0.010	0.0046	0.0016	0.0005
Closeness centrality	0.60	0.57	0.54	0.51
Eigenvector centrality	0.068	0.054	0.040	0.030
Clustering coefficient	0.63	0.65	0.74	0.79

Core Nodes and Central Structure



01

Core Hub Identification

Based on comprehensive calculations of degree, betweenness, closeness, and eigenvector centrality, Captain America ranks first in all indicators and is the absolute core hub of the Marvel network.

02

Top Three Core Characters

The top three core characters are Captain America, Wolverine, and Spider-Man, acting as important bridges across different teams and eras.

03

Functional Features of Core Nodes

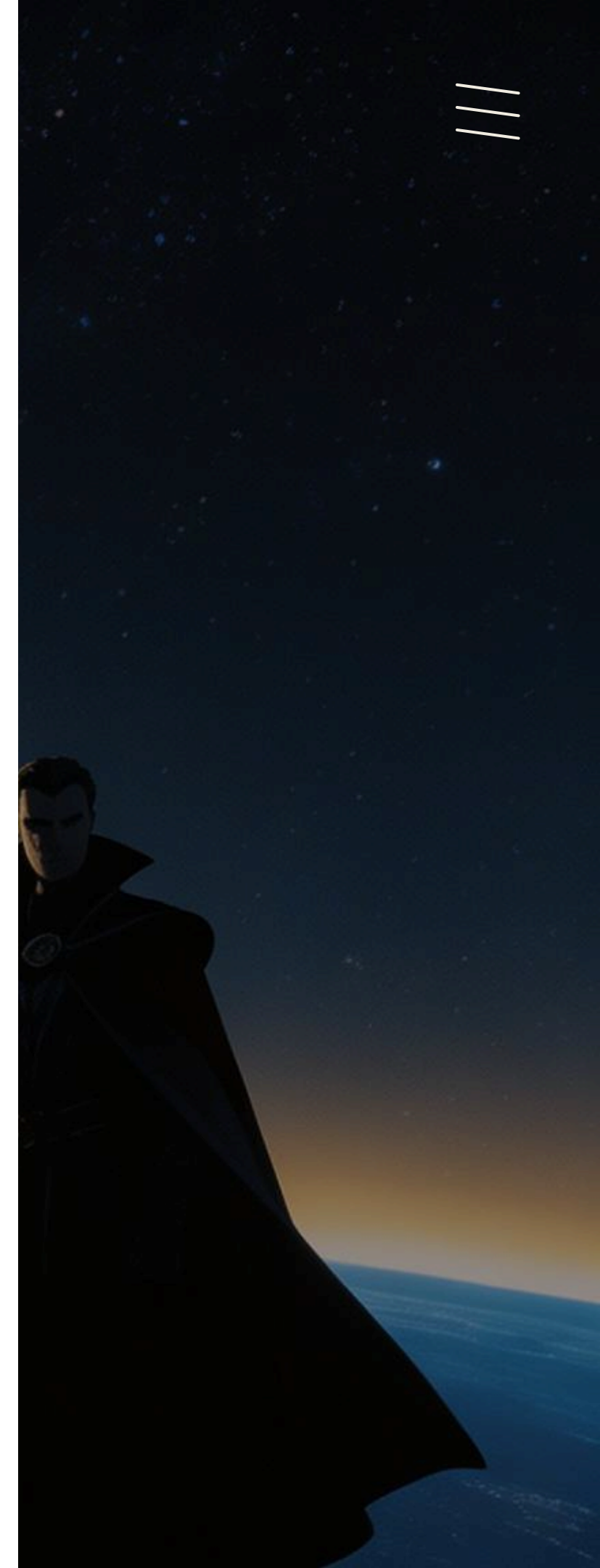
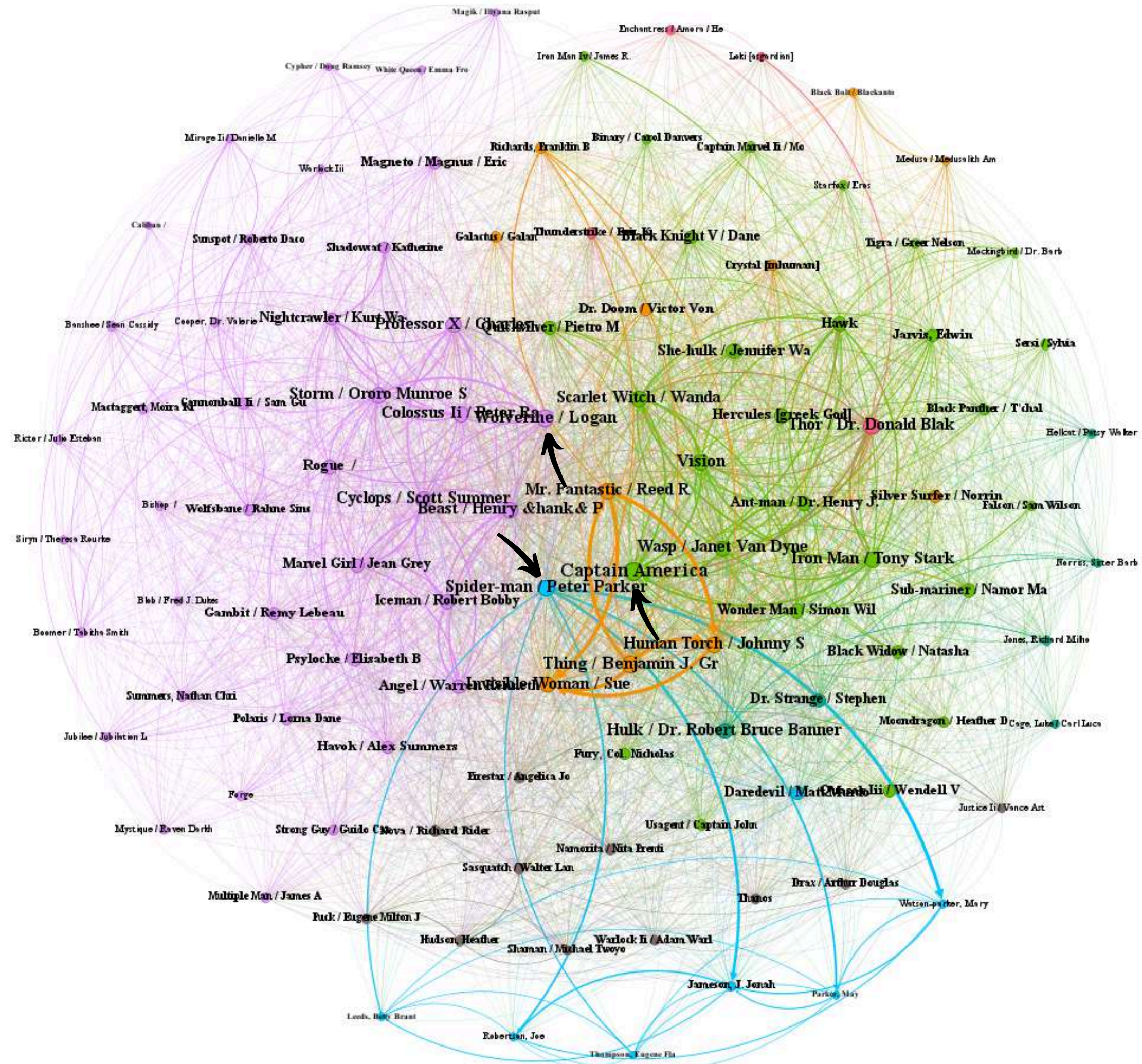
Core nodes have high connectivity and frequent cross-community interactions, connecting the two core communities of the Avengers and the X-Men.

Core Nodes and Central Structure

Rank	Character	Era	Gender	Centrality	Core Team
1	Captain America	Before 1961	Male	0.791	Avengers, Invaders, Illuminati
2	Wolverine / Logan	Bronze Age	Male	0.675	X-Men, Avengers, X-Force, Alpha Flight
3	Spider-Man / Peter Parker	Silver Age	Male	0.669	Avengers, Fantastic Four, Future Foundation, New Warriors
4	The Thing / Benjamin J. Gr	Before 1961	Male	0.656	Fantastic Four, Avengers, Guardians of the Galaxy, Heroes for Hire
5	Vision	Silver Age	Male	0.653	Avengers, West Coast Avengers, Illuminati
6	Beast / Henry “Hank” P	Silver Age	Male	0.629	X-Men, Avengers, Defenders, X-Factor, Illuminati
7	Thor / Dr. Donald Blak	Silver Age	Male	0.629	Avengers, Guardians of the Galaxy, Revengers, Asgardians
8	Iron Man / Tony Stark	Silver Age	Male	0.623	Avengers, Illuminati, Guardians of the Galaxy
9	Mr. Fantastic / Reed R	Before 1961	Male	0.607	Fantastic Four, Illuminati, Future Foundation
10	Cyclops / Scott Summers	Silver Age	Male	0.604	X-Men, X-Factor, Phoenix Force

Metric	1st Place	2nd Place	3rd Place
Centrality	Captain America (0.79)	Wolverine (0.67)	Spider-Man (0.67)
Betweenness centrality	Captain America (0.067)	Spider-Man (0.062)	Wolverine (0.035)
Closeness centrality	Captain America (0.83)	Wolverine (0.75)	Spider-Man (0.75)
Eigenvector centrality	Captain America (0.139)	Wolverine (0.131)	The Thing (0.130)





Node Attributes and Network Position

01 Gender:

Female characters are absent from the top 10 core centrality rankings, with only a few female characters on the eigenvector centrality and clustering coefficient lists.

02 Race:

The proportion of white characters decreases year by year, racial diversity improves in the modern era, and alien/non-human characters increase significantly.

03 Alignment:

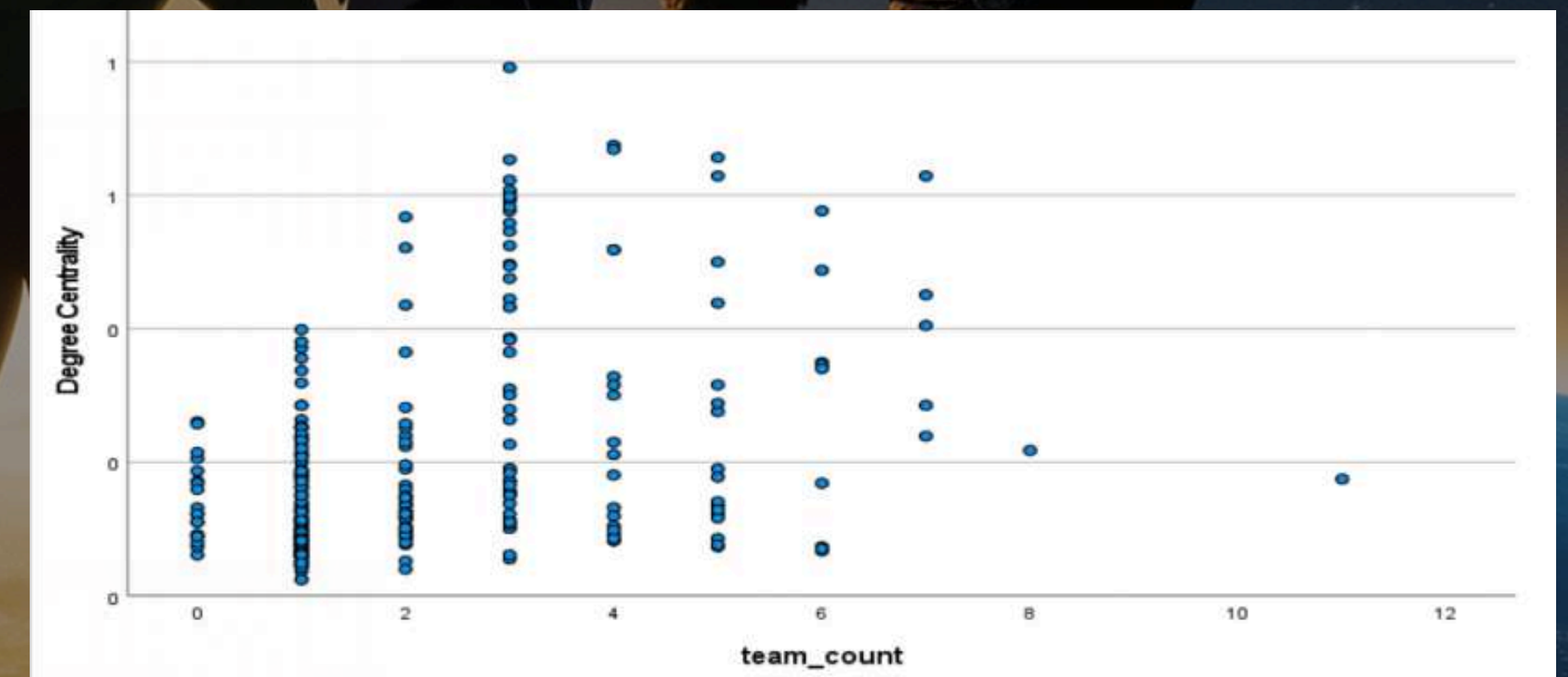
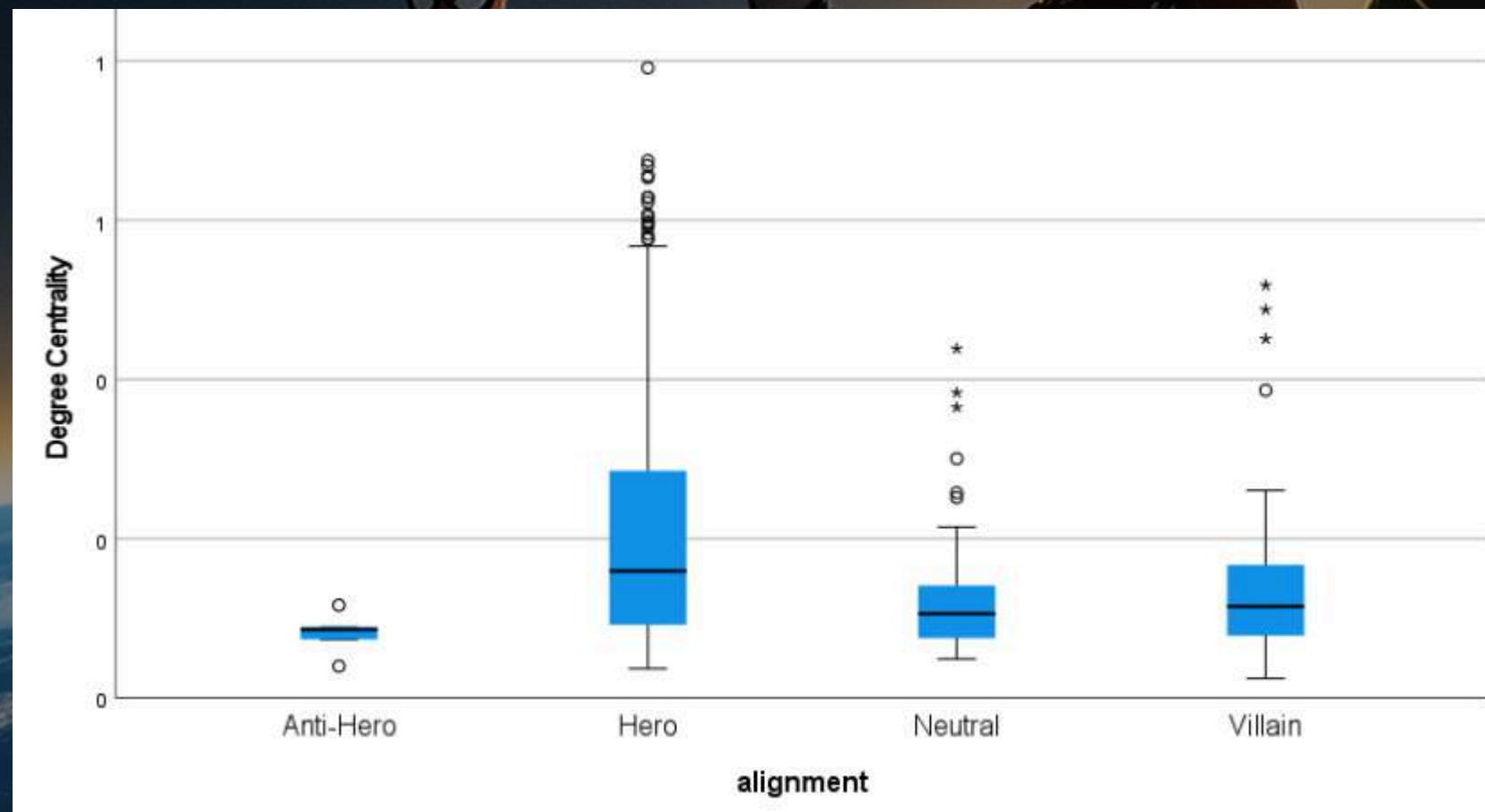
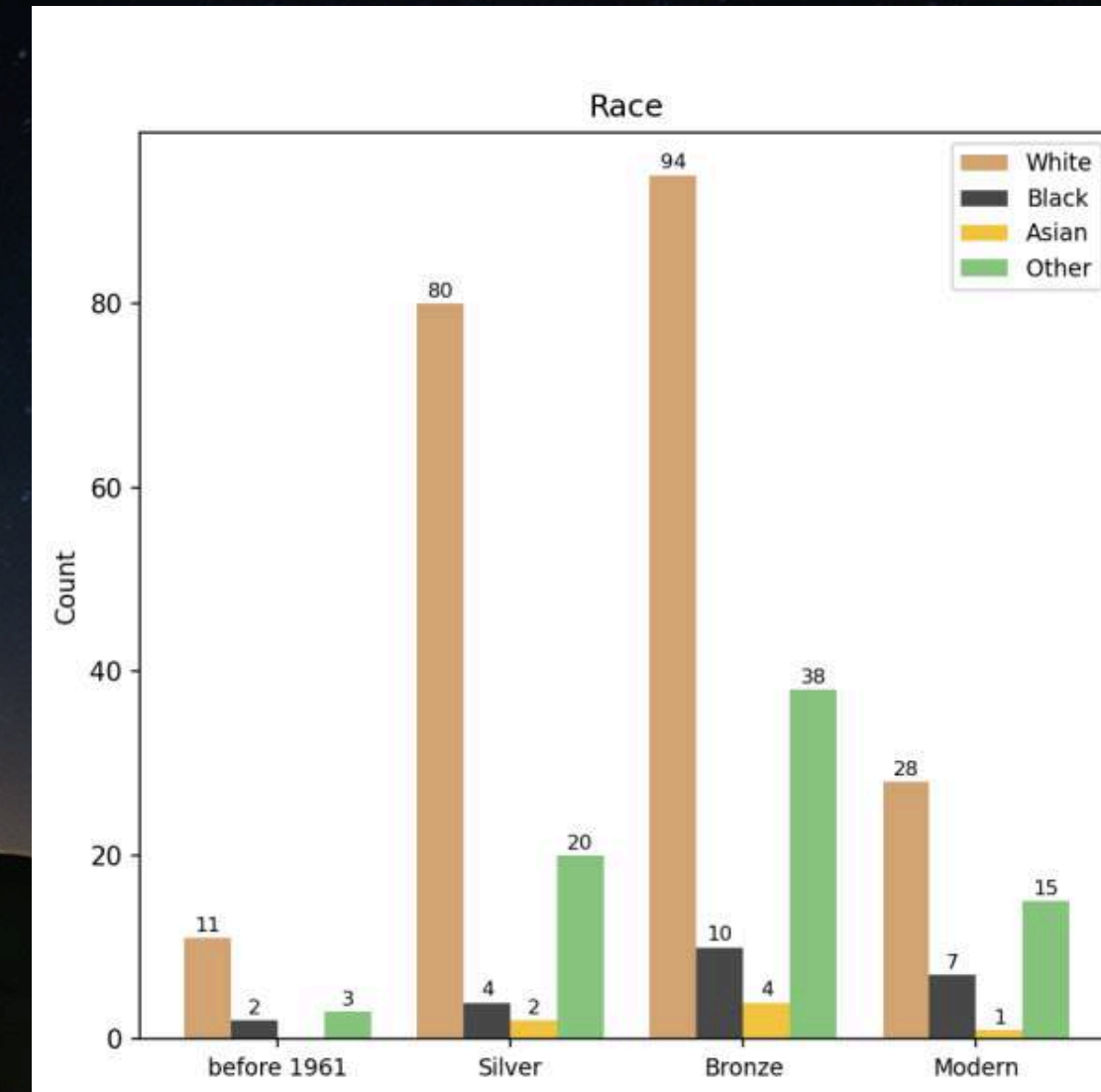
Hero characters have the highest centrality, anti-heroes the lowest, and villains and neutral characters are at a medium level.

04 Team participation:

The number of teams a character joins is significantly positively correlated with centrality, and cross-team characters have stronger network influence.

Node Attributes and Network Position

Metric	Number of Female Characters in the Top 10	Specific Characters
Degree centrality	0	None
Betweenness centrality	0	None
Closeness centrality	0	None
Eigenvector centrality	2	Invisible Woman (8th), Scarlet Witch (10th)
Clustering coefficient	3	Volcana, Asp II, Black Mamba



Regression Analysis Results

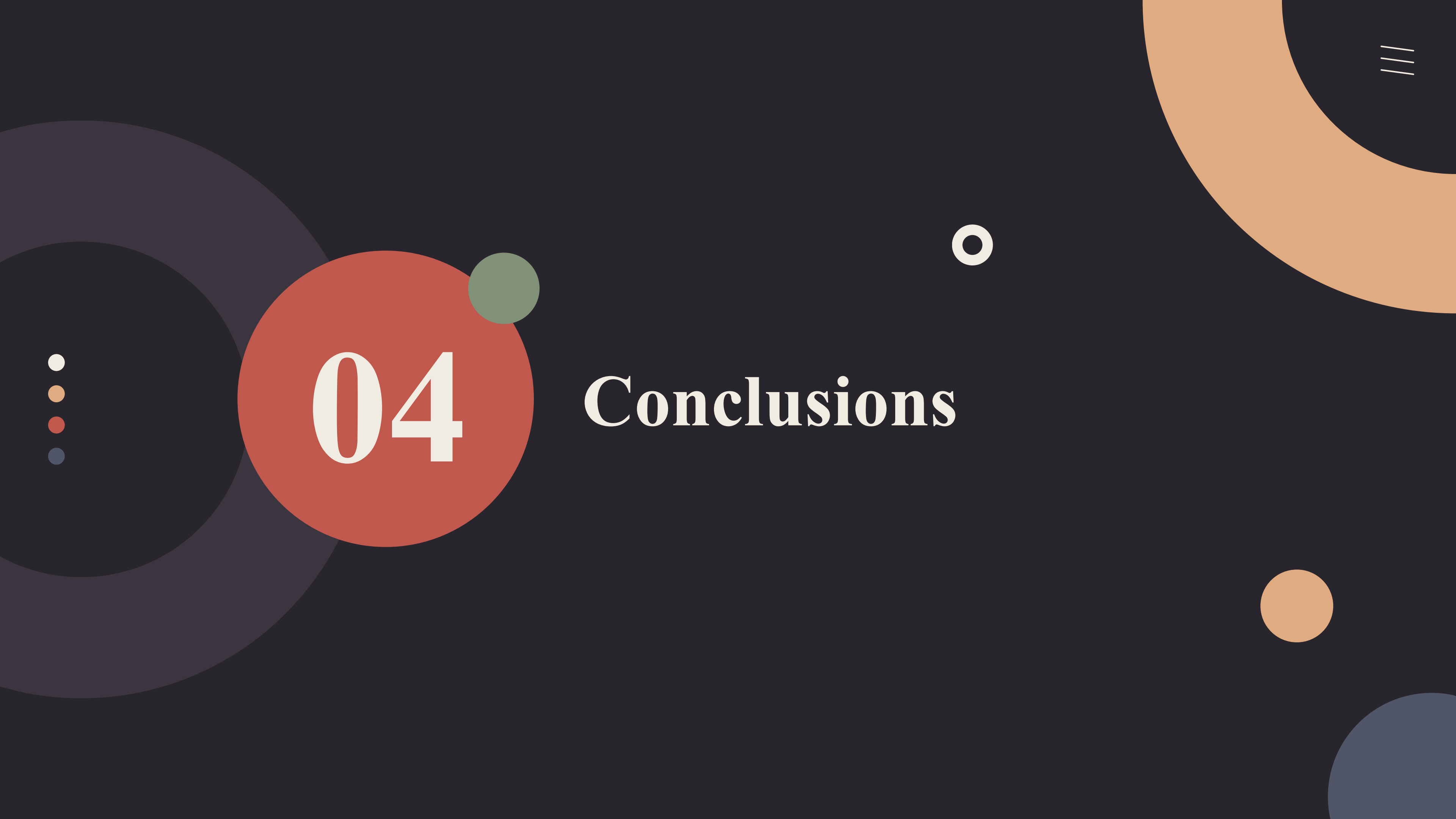
Coefficients ^a					
模型		未标准化系数		标准化系数	
		B	标准错误	Beta	t
1	(常量)	.093	.049		1.920
	team_count	.036	.004	.401	8.225
	gender_num	.027	.016	.083	1.701
	alignment_num	-.040	.008	-.234	-4.803
	race_num	.022	.010	.107	2.187

a. 因变量：Degree Centrality



- 01
- The regression model is significant ($F=25.63$, $p<0.001$), Adjusted $R^2=0.232$, explaining 23.2% of the variance in degree centrality.
- 02
- Team participation is the strongest significant factor affecting character centrality ($\beta=0.401$, $p<0.001$).
- 03
- Alignment and race have significant effects, while gender has no statistically significant effect on centrality.





04

Conclusions

Research Conclusions

01

Network Structure Characteristics

The Marvel character network has small-world and core-periphery structure, with modularity $Q=0.435$, divided into 7 stable communities, and the Avengers and X-Men as dual cores.

02

Character Evolution Pattern

Early classic characters form the network core, modern characters show small-group dynamics, and cross-era connections are weakened.

Research Conclusions

04

Core Resource Distribution

Core resources are highly concentrated in classic white male heroes, while female and minority characters are still on the periphery.

03

Determinants of Network Core Status

Team participation determines a character's network core status more than demographic attributes such as gender and race.

Implications, Limitations and Future Research

Research Implications:

- Marvel builds a narrative system with core hub characters to ensure IP linkage and scalability.
- Diversity reform only stays at the level of character quantity, without essential optimization of network structure.
- Social network analysis can be effectively applied to the narrative and operation research of film and television IPs.

Research Limitations:

- Only highly active characters are selected, and peripheral characters are insufficiently representative.
- The network is only built based on co-occurrence relationships, without distinguishing the specific nature of character interactions.
- The era division adopts industry consensus standards with certain subjectivity.





Thanks For Watching!



Group 3